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Arbitrary Waveforms using a Tri-State Transmit Pulser

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***Abstract*— Conventional hardware for arbitrary waveform formation uses Digital-to-Analog Converters (DAC) and linear amplifiers. Proposed here are methods for encoding a tri-state pulser to approximate arbitrary waveforms. Algorithms are evaluated by simulation and experiment using the Verasonics, Inc. Vantage research ultrasound system, which transmits with 4 ns edge resolution. Two problems were considered: 1) matching the waveform produced by an ideal DAC given a particular transducer, and 2) compensating for the impulse response of a given transducer to reproduce a signal. The problems require the transducer impulse response in one-way and two-way propagation geometries, estimated here in water using a hydrophone and the ATL L7-4 transducer. Within measured nonlinear propagation effects, error in these four scenarios for an LFM waveform was consistent with simulated performance. In another test, a Gaussian pulse is encoded for transducer compensation, and compared to equivalent-bandwidth monopulse excitation in planewave imaging on a phantom. Here, a pin-trailing pedestal artifact is reduced by 3 dB over a span equal to the peak resolution (FWHM).**

Keywords- waveform synthesis; transducer compensation; RF amplifier; linear equalization; impulse response estimation; deconvolution; least squares signal approximation; 1-bit conversion; delta modulation

I. Introduction

We present methods for encoding waveforms into sequences suitable for a tri-state RF ultrasonic transmitter, under various fidelity criteria. The encoding concept is demonstrated on the Vantage Ultrasound System (Verasonics, Inc. Redmond, WA, USA). In a prior work, the authors of [1] demonstrate an example of time-reversal pulse compression using a binary transmitter encoded with a zero-crossing -based strategy. While they report performance that compares favorably to a digital-to-analog converter (DAC) -based transmitter, other applications may have more stringent performance requirements.

A. Contribution

The encoding methods presented require knowledge of the transducer element's impulse response (IR). We describe an impulse response estimation method, and use the results to introduce encoders for tri-state pulser sequences. The encoders are based on constrained deconvolution concepts from communications science known as "equalizers" [2], combined with a hybrid pulse-width modulation (PWM) signaling and quantization scheme.

In four acoustic water-tank test scenarios using an ATL/Philips L7-4 transducer, a linear frequency modulated (LFM) signal is encoded and transmitted. The method achieved normalized RMS error (NRMSE) fidelity consistent with simulated performance, within quantified nonlinear propagation effects. In another test, a Gaussian pulse (GP) is encoded for transducer compensation, and compared to equivalent-bandwidth monopulse excitation in planewave (PW) imaging on a resolution phantom. Pin-trailing "pedestal" artifacts are reduced by 3 dB over the coincident profile spanning full-width half-maximum (FWHM).

B. Outline

A brief description of the transmitter operation is given. The usage models dictating mathematics of the problem are described. The estimation and encoding algorithms are then introduced. The experimental approach is documented, and results are then discussed, with attention given to interpreting performance of both the tri-state and DAC-based transmitter architectures under nonlinear propagation.

II. Background

A. Transmitter Description

The Vantage Ultrasound System transmitter allows specification of arbitrary sequences of the three voltage levels [+V,0,-V] at 4 nS clock intervals. Each acquisition event may have sequences unique to each transmit element. The sequences may be of arbitrary length, subject to memory limitations and power supply capacity. One restriction is a 3-clock minimum state dwell required to enter a given voltage level state. Another is that the realized voltage is approximately the 5-clock running average of the voltage of the achieved state.

The Vantage system uses several different methods to specify a transmitted waveform, including a parametric method; an apodized FM specification; and a run-length encoding denoted "pulse code", used here exclusively.

B. Encoder Usage Models

The fidelity metric employed by the encoder is a design choice that depends on the usage model of the application. NRMSE metrics considered here measure (1) closeness of the reference waveform to acoustic pressure, and (2) closeness of the reference waveform's predicted acoustic pressure to that achieved. We denote these the one-way DAC synthesis and one-way transducer compensation problems, respectively. Their two-way versions compare the reference to received data, rather than acoustic pressure.

III. Algorithm

The presented method of waveform encoding is motivated by the concept of symbol "equalization" when communicating through bandlimited channels [2]. This concept is generalized here to the symbol-block case; this means the entire sequence of transmit pulses (for a channel) are optimized together, rather than serially as individual pulses. This problem of symbol inference might be interpreted as a deconvolution constrained to a discrete set of inputs. Another important difference between the communication problem and the use of the equalization concept here, is the performance metric. In the communication problem, a true symbol sequence exists against which performance can be measured, in terms of symbol error rate. In the encoding problem, no "true" symbol sequence exists, and the objective is only to fit the acoustic pressure generated by the transducer (or receiver data) to the design waveform, subject to allowed inputs (symbols).

A. Process Architecture and Operation

The components and operation of the encoding process are shown in the architecture diagram of Fig. 1. These components include IR estimation, Symbol Set definition, Symbol Set calibration, Symbol Quantization, and Equalization. The input to the process is the reference design waveform, represented by high precision samples of an analog signal or function. The output of the process is a tri-state pulse sequence suitable for control of the transmitter pulser circuitry.

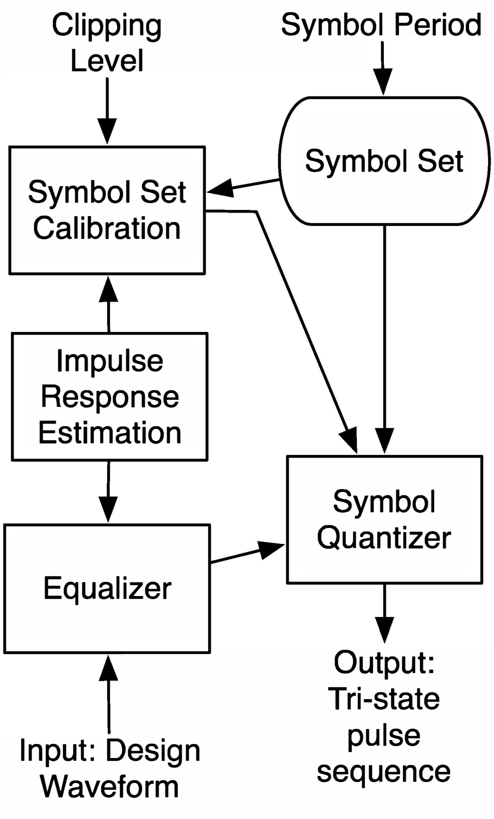

Figure 1 Arbitrary waveform synthesis process architecture.

B. Impulse Response Estimation

Initially, the architecture requires that IR estimation be conducted for each transmit element, according to the signal path determined by the usage model. The impulse response estimation is formulated as convolution implemented in a linear statistical model and solved by least-squares theory [3]. This technique is demonstrated in prior work on underwater acoustic data at sonar frequencies [4]. Here, the model has as input a known sequence of transmitted pulses collected in a vector $\mathbf{q}$, and formed into a Toeplitz matrix [3]. Unknown parameters comprising the impulse response are represented by the vector $\mathbf{h} = [h(1),\ldots,h(L)]^T$, in

$$\mathbf{Y} = \mathbf{Q}\,\mathbf{h} + \mathbf{e} \tag{1}$$

where the modeling error is represented by $\mathbf{e}$, the measurement is $\mathbf{Y} = [y(1),\ldots,y(N+L-1)]^T$, and the model data is the Toeplitz matrix $\mathbf{Q}$=T($\mathbf{q}$) of the zero-padded vector $\mathbf{q}$, defined as

$$\mathbf{Q} = \begin{bmatrix} q(1) & 0 & \cdots & 0 \\ \vdots & q(1) & & \vdots \\ q(N) & \vdots & & \\ 0 & q(N) & \ddots & 0 \\ & 0 & \ddots & q(1) \\ \vdots & & & \vdots \\ 0 & \cdots & 0 & q(N) \end{bmatrix}. \tag{2}$$

One means of solving (1) is by the pseudoinverse [3], which gives the estimate of impulse response of vector $\mathbf{h}$ as

$$\hat{\mathbf{h}} = \mathbf{Q}^{+}\mathbf{Y} \,. \tag{3}$$

C. Parameter Selection

Parameters needed for the algorithm include clipping level and symbol period D. The symbol period is typically based on the nominal transducer center frequency. Each symbol is comprised of several transmit clock periods and therefore several tri-state voltage instances. A typical choice of symbol period corresponds to 1/4 the nominal center frequency. Additionally, the symbol period selection determines the number of PWM levels available. The hypothetical twelve-clock symbol period permits 25 PWM levels, including the zero-voltage level. The two consequences of a symbol period choice form a design tradeoff; a larger symbol period provides more PWM levels, at the cost of reduced symbol rate. In the experiments here, the symbol set is formed from a combination of PWM and frequency modulation (not shown).

The clipping level is a companion parameter to the symbol set. It adjusts the overall level, with respect to the equalizer output, of the calibration gains when they are applied in the symbol quantizer. In this way the range of the equalizer output is fit within the quantizer window. The optimum clipping level is found empirically for each reference design waveform encoded, so the sum-square error between the reference and synthesized waveforms is minimized.

D. Symbol Set Calibration

The symbol set calibration determines a gain mapping between each symbol in the constellation, and its equivalently-weighted Dirac impulse, as seen at the output of the transducer. This is achieved as the least-squares solution for the gain variable $g(k)$ in

$$\mathbf{S}_k = g(k)\,\mathbf{S}_0 + \mathbf{e} \tag{4}$$

where the vector $\mathbf{S}_k$ represents the response of the impulse response model when convolved with the k-th symbol, and vector $\mathbf{S}_0$ is the response to the prototype reference symbol, typically chosen to be the largest-bandwidth symbol.

E. Equalization

The equalizer component deconvolves the reference design waveform against a model derived from the usage model's signal path. The output of the equalizer comprises a sequence vector **r** of "soft" symbols, which represent continuously-valued Dirac impulse weights. When these are convolved with the IR model, the result approximates the specified design waveform. In analogy to the IR estimation, a modified Toeplitz matrix formed from the estimated IR vector is used as part of a linear statistical model describing convolution, so that the vector of reference design waveform samples

$$\mathbf{W} = [w(1),\ldots,w(L+P\text{-}1)]^{\mathrm{T}} \tag{5}$$

is interpreted as the measurement in

$$\mathbf{W} = \mathbf{H}_d\,\mathbf{r} + \mathbf{e} \tag{6}$$

where the model matrix $\mathbf{H}_d$ is the column-decimation of the Toeplitz matrix T(**h**) associated with zero-padded IR vector **h**,

$$\mathbf{H}_d = \begin{bmatrix} h(1) & 0 & \cdots & & 0 \\ \vdots & & & & \vdots \\ h(L) & \ddots & h(1) & & \\ 0 & & \ddots & & 0 \\ & 0 & h(L) & \ddots & h(1) \\ \vdots & & & & \vdots \\ 0 & & \cdots & 0 & h(L) \end{bmatrix}. \tag{7}$$

The decimation factor by which a subset of columns of $\mathbf{H}_d$ is retained from the standard Toeplitz matrix corresponds to the symbol period D. For example, a symbol period of 12 transmit clocks means every 12-th column of the standard Toeplitz matrix **H**=T(**h**) is retained as a column of $\mathbf{H}_d$. The unknown parameter vector is the set of soft symbols in the vector **r**, as

$$\mathbf{r} = [r(1),\ldots,r(L/D+P\text{-}1)]^{\mathrm{T}}\,. \tag{8}$$

The solution for parameter **r** is found by the pseudoinverse as

$$\hat{\mathbf{r}} = \mathbf{H}_d^{+}\mathbf{W}\,. \tag{9}$$

For some design waveforms, an iterative extension (here labeled as the "conditional equalizer") may give better performance. In this method, the equalizer output is quantized according to the symbol quantization step in the sequel. The resulting symbol sequence **p** is then applied to the impulse response in the new model notated as

$$\mathbf{W} = \mathbf{H}_d\mathbf{p} + \mathbf{e}\,. \tag{10}$$

Then, sequentially for each element of the vector **p**, the element is replaced with the symbol from the symbol set that gives the least squares residual fitting error. This is determined by a search of the symbol set, and repeated for all elements of **p**. If the incumbent element values of **p** give the lowest error, the process stops; otherwise, it is repeated. The symbol set used in this process may have a different symbol period than that used initially, adjusting sizes and decimation factors.

F. Symbol Quantization

The symbol quantizer component chooses, for each soft symbol in the equalizer output **r**, the symbol closest in terms of the gain mapping determined in the calibration step. Thus, it chooses the $g(k)$ closest among each k to the soft symbol being mapped. The chosen symbols are then converted into their constituent trinary pulse sequences and concatenated to a single tri-state sequence.

IV. Experimental Results

A water tank experiment was conducted to validate performance of the IR estimation and tri-state encoding process.

A. *Two-way Experiment Configuration*

The experiment consisted of a Philips L7-4 Transducer fixed in a water tank, and pointed directly at an acrylic block of 5.08cm thickness. The Verasonics Vantage acquisition system is connected to the transducer.

B. *Impulse Response Estimation Experiment*

For transducer impulse response identification, a single element channel is chosen for transmission and reception. The acrylic block acts as an acoustic mirror. Eight distinct pulse trains are transmitted as probing sequences in separate acquisition events, from which the IR estimate is computed.

C. *LFM Synthesis Test*

Encoding is demonstrated on a 10 ▪ s duration LFM signal with a Taylor weighted envelope. The frequency ranged from 3.5 MHz to 6.5 MHz. Table 1 shows the NRMSE between the reference waveform and the waveform after receiver filtering.

D. *Imaging with Equalized Gaussian Pulse*

In this test, a 60% bandwidth GP is encoded for transducer compensation, and compared to an equivalent-bandwidth monopulse in PW imaging on a 0.7 dB/cm/MHz resolution phantom. Seven PW angles are used. The monopulse is transmitted at 5.0 V, while the GP is transmitted at 6.24 V, to equate their beamformed intensities. Fig. 2 shows profiles of reconstructed image slices of the two pulses at a wire target 2 cm deep. Here, pin-trailing pedestal artifacts are reduced by 3 dB over a span equal to the peak resolution.

E. *Nonlinearity Test*

The residual sum-of-squares from the impulse response estimation is much higher than quiescent noise variance would predict, suggesting a nonlinearity in the signal chain. Tests with a 10-bit DAC waveform generator driving a 50W linear amplifier into the transducer implicate the water as the source of nonlinearity, under practical transmit voltages. Using as test signals a 10 cycle 5 MHz sinusoid (DAC) and square wave (Vantage), visibly periodic distortion is quantified by a second harmonic level of -25.5 dB at 8V. At higher voltages, distortion increased uniformly and identically for the DAC and Vantage system. At lower voltages, single-cycle bursts compared in a pulse-inversion scenario sensitive to voltage imbalance, show a distortion of -24.2 dB at 5V for the Vantage.

TABLE I. LFM TANK TESTS

NRMSE (dB)	Experiment Conditions		
	Usage Model	***Geometry***	***TX Voltage***
-23.8	Transducer compensation	one-way	8.0 V
-24.1	DAC synthesis	one-way	8.0 V
-22.5	Transducer compensation	two-way	5.0 V
-21.6	DAC synthesis	two-way	5.0 V

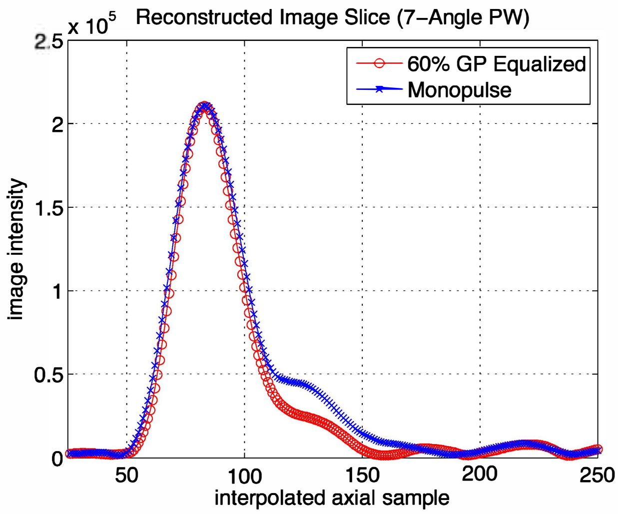

Figure 2 Image slice with Gaussian pulse and monopulse.

V. SUMMARY

We described a method for encoding waveforms of continuous-valued specification into a tri-state transmitter sequence, and presented measured acoustic performance on LFM and GP signals respectively by NRMSE and sidelobe artifact intensity. While the values in Table 1 are approximately 3 dB higher than simulation (not shown), the achievable measured NRMSE is likely limited by the nonlinearities as measured in IV-E. To explain this, we predict the in-band error due to nonlinearity in the 8 V LFM case to be roughly -25.5 dB in the one-way experiment, since the LFM has uniform spectral intensity across the large part of the transducer bandwidth. Combining this noise with intrinsic algorithm error (-27.7 dB by simulation), the result assuming uncorrelatedness would be a total error of -23.5 dB. Thus, the results of the LFM tests are not inconsistant with simulation, however suggesting the need for experiments not subject to the water medium nonlinear propagation effects.

Some content presented here is patent pending by USPTO application number 61856488.

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